Darshan Sonagara

Lead Data Scientist, Search Retrieval · Target · Minneapolis, MN

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SUMMARY

Lead Data Scientist who builds machine learning systems for search, information retrieval, recommendation, and trust and safety, serving millions of people every day. 9+ years taking ML from research to production, spanning NLP, knowledge graphs, real-time fraud detection and misinformation detection. Built Target's first semantic search channel and led its launch as a hybrid system serving all Target.com and app search; first-named inventor on its patent application and first author of the paper describing it. Research cited 331 times (Google Scholar, Oct 2026); JASIST Top Cited Article award, two consecutive years.

EXPERIENCE

Target · Minneapolis, MN — Apr 2023 – present

Lead Data Scientist, Search Retrieval (Oct 2026 – present) · Senior Data Scientist (Apr 2023 – Oct 2026)

Semantic and hybrid product search · *dual encoders, contrastive learning, PyTorch, AlloyDB/ScaNN, Solr, A/B testing*

- Built Target's first semantic retrieval channel from scratch and led its launch alongside lexical search (Solr); reached 100% of Target.com and app traffic in Feb 2025 after 7+ A/B iterations.
- Selected and fine-tuned the embedding model (e5-small-v2) on ~20M query–product pairs labeled from engagement (purchase > add-to-cart > click), with exposure- and similarity-based hard negatives and a bidirectional contrastive loss; designed field-marker tokens and field masking to combine product attributes robustly.
- Built precision control using NER and a query classifier to filter vector results by item type, category, brand, gender and size: precision@1 0.890 → 0.929, precision@10 0.836 → 0.885.
- Replaced reciprocal rank fusion with weighted result interleaving after A/B tests showed RRF hurt low-overlap result lists.
- Four-week online A/B results vs lexical-only search: CTR +0.97%, order conversion +0.98%, demand per visitor +1.10%, ~50% fewer zero-result searches.
- Owned the production query and item embedding services (p99 < 50 ms on CPU with caching and adaptive batching) as sole point of contact through peak seasons with zero incidents; wrote the runbook other teams adopted as a template.
- Embedding models adopted by Target's Ads, Content Understanding and Query Understanding teams.

LLM-based relevance labeling and precision control · *agentic frameworks, LLM fine-tuning, knowledge distillation, cross-encoders, PySpark*

- Designed a three-stage system: an agentic, LLM-driven labeling pipeline that produces relevance labels with reasoning; an in-house LLM fine-tuned on that reasoning data for relevance prediction; and a lightweight cross-encoder or shared bi-encoder student for online precision control.

Personalized retrieval (2025) · *two-tower models, attention, PyTorch, Vertex AI, Redis*

- Moved personalization into the retrieval layer: designed user, query and item towers, with query-guided attention over real-time, short-term and long-term behavior (queries, clicks, add-to-carts, purchases) producing a user-aware query context vector for a dedicated personalized retrieval channel; the winning variant improved Recall@K, NDCG and MRR offline.
- Built multi-GPU training on Vertex AI, PySpark pipelines for user-history training data, a Redis Sentinel feature store with sub-millisecond embedding reads, and an offline evaluation framework.

Goldman Sachs · New York, NY — Feb 2019 – Apr 2023

Vice President · also Summer Intern, May – Aug 2018

Enterprise search for compliance surveillance (2022 – 2023) · *BM25, dense retrieval, result fusion, Elasticsearch*

- Led design and training of in-house retrieval models for high-recall search over sensitive, access-controlled data on the platform built for the Compliance division and later adopted firm-wide; led multi-channel fusion of lexical and semantic results.
- Contributed to the sharding design and to entity extraction and tagging that links documents to the firm's knowledge graph.

Real-time fraud detection using a knowledge graph (2020 – 2022) · *graph features, fraud models, graph databases*

- Led feature engineering on graph-derived signals and the fraud models for the Apple Card program, screening new applications and live transactions in real time; contributed to the knowledge graph's design and infrastructure.

Search, recommendation and document intelligence (2019 – 2022) · *TensorFlow, PyTorch, Hugging Face, Elasticsearch, knowledge graphs*

- Built neural ranking for search over finance, news and broker research, and a two-tower recommendation component for personalized article suggestions, evaluated with recall, precision, NDCG and A/B tests.
- Built named-entity disambiguation that links entities in financial text to the knowledge graph, combining a Bayesian linking algorithm with learned entity embeddings.
- Built a distributed enrichment pipeline (industry classification, entity extraction, sentiment, language detection) and a financial document classification and field-extraction system.

Rutgers University, Behavioral Informatics Lab · New Brunswick, NJ — Jun 2017 – Dec 2018

Research Assistant to Prof. Vivek K. Singh

- Research assistant on NSF-funded cyberbullying detection research (Award IIS-1464287); designed the multimodal feature set for Instagram cyberbullying detection and led labeling of a new dataset.
- Multimodal fake-news detection (JASIST 2021): identified the text and visual features that separate fake from credible news, designed a scalable extraction pipeline, and built the multimodal models (95.18% accuracy vs 86.24% text-only).
- Linguistic fake-news detection (SBP-BRiMS 2017): co-built the classifier and two public credible-news datasets.

Tata Consultancy Services · Kolkata, India — Mar 2015 – Dec 2016

Assistant System Engineer

- Built mainframe-to-Hadoop data migration and moved CPU-heavy batch jobs to Hadoop and Hive; built a product-review sentiment analyzer.

PUBLICATIONS

D. Sonagara, Q. Zhang, A. Singh, A. Li. **Retail Product Search: A Practical Approach at Target.** arXiv:2609.31498, 2026.

V. K. Singh, I. Ghosh, D. Sonagara. **Detecting fake news stories via multimodal analysis.** JASIST 72(1): 3–17, 2021. DOI 10.1002/asi.24359. 191 citations.

V. K. Singh, R. Dasgupta, D. Sonagara, K. Raman, I. Ghosh. **Automated fake news detection using linguistic analysis and machine learning.** SBP-BRiMS Grand Data Challenge, 2017. 98 citations. Name printed as "D. Sonagra".

D. Sonagara, S. Badheka. **Comparison of basic clustering algorithms.** IJCSMC 3(10): 58–61, 2014.

PATENT, TALKS AND RECOGNITION

- **Patent application:** hybrid search system with adaptive precision control and weighted result interleaving for product retrieval. U.S. provisional patent application filed Apr 2026, assigned to Target; first-named inventor.
- **Award:** JASIST Top Cited Article award, 2020–21 and 2021–22.
- **Talk and poster:** SBP-BRiMS 2017 Grand Data Challenge, George Washington University, Washington, DC (presented by Prof. Vivek K. Singh).
- **Coverage of the work:** Google Cloud blog on Target's hybrid search system, Sep 2025.

EDUCATION

M.S. Computer Science, Rutgers University, New Brunswick, NJ · GPA 3.68/4.0 — 2019

B.E. Information Technology, G H Patel College of Engineering & Technology, Anand, India · GPA 3.70/4.0 — 2015

SKILLS

Search and IR	Hybrid search, dense and lexical retrieval, dual encoders, contrastive learning, hard-negative mining, ANN indexes, result fusion and interleaving, precision control, query understanding, learning to rank
LLMs and NLP	LLM fine-tuning, knowledge distillation, cross-encoders, agentic frameworks, named-entity recognition and disambiguation, text classification
Other ML	Recommender systems, two-tower models, attention over user behavior, knowledge graphs, fraud detection, multimodal (text and image) models
Engineering	Python, PyTorch, TensorFlow, Transformers, PySpark, Solr, Elasticsearch, AlloyDB, Redis, Kubernetes, KServe, Vertex AI, GCP, Hadoop, A/B testing